

Rakin Ahmed

MECHANICAL ENGINEER | ADDITIVE MANUFACTURING & POLYMER PROCESS R&D

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Mechanical engineer and Ph.D. candidate (December 2026) specializing in additive manufacturing, polymer process development, and materials characterization. Inventor of a patent-pending in-process annealing printhead that recovers up to about 92 percent of bulk strength in 3D-printed parts. Combines four years of industry design experience across fuel-storage terminals, piping systems, and product development with a research record of 13 peer-reviewed publications and a full experimental-to-modeling workflow. Seeking R&D, process-development, and materials-engineering roles in industry or a postdoctoral setting.

PROFESSIONAL EXPERIENCE

Graduate Research Assistant, **University of Texas at Arlington**

Aug 2022 – Present

Additive Manufacturing and Polymer Process Research, MAE Department (Advisor: Dr. Robert M. Taylor)

- Invented and prototyped a near-contact in-process annealing printhead, now in patent filing, that raises interlayer bond strength to about 86 percent of bulk PLA on average and a best replicate near 92 percent by adding programmed interlayer cooling through modified G-code.
- Designed and executed full-factorial designed experiments that isolated crystallization-arrested healing as the controlling bond mechanism, quantified by differential scanning calorimetry (15.5 versus 3.1 percent crystallinity, $p = 0.003$).
- Developed a second-generation modular printhead that cuts the custom part count from five to three and integrates off-the-shelf heating elements, improving reliability and lowering machining cost.
- Built coupled non-isothermal reptation and porosity-strength models in Python using WLF shift factors to predict bond development directly from a part's measured thermal history.
- Engineered more than 40 thermal-signature features from FLIR thermography and trained PCA-based machine-learning models (XGBoost, CatBoost, ANN) reaching 86.7 percent balanced accuracy and 0.91 AUC for non-destructive toughness classification.
- Established a multimodal characterization workflow spanning SEM, micro-CT, DSC/STA, FTIR, Raman, DIC, and acoustic emission to connect process, structure, and mechanical performance.
- Extended the crystallization-controlled bonding framework from solid to foaming PLA and validated it at component scale on thin-walled stiffened box beams.
- Authored a 48-page critical review on design for additive manufacturing of thin-walled and lattice structures; contributed to 13 peer-reviewed publications and serve as a reviewer for four journals.

Graduate Teaching Assistant, **University of Texas at Arlington**

Aug 2022 – Present

- Lead laboratory instruction in Experimental Methods and Measurement and in Additive Manufacturing, maintaining a fleet of more than 20 FFF printers and training students on printing, tensile testing, and FaroArm reverse engineering.
- Mentored student teams in the UTA 3D-Printed Aircraft Competition and supervised an undergraduate researcher, earning the UTA I-ENGAGE Undergraduate Student Mentorship Award.

Executive, Business Development, **Omera Petroleum Ltd.**

Feb 2021 – Aug 2022

Dhaka, Bangladesh

- Delivered operational audits of LPG bottling plants and headquarters that drove process improvements reducing production costs by about 30 percent.
- Structured and helped close a strategic partnership appointing Rock Energy as sole marketer of Caltex-branded lubricants for Chevron Lubricants Lanka.
- Conducted feasibility studies for new energy-sector ventures and supported the rollout of approved LPG and petroleum projects.

Senior Design Engineer, **Trigger Automation & Engineering Solutions Ltd.**

Sep 2019 – Jan 2021

Dhaka, Bangladesh

- Designed structural components and piping for a 45,300 cubic meter (1.6 million cubic foot) Heavy Fuel Oil storage terminal of ten fixed-cone welded tanks in compliance with API 650.
- Engineered NFPA 11 fire-protection and foam-suppression systems for a methanol tank farm, and designed and supervised fabrication of a 40,000 L underground diesel storage tank to CAN/ULC-S603.
- Produced plant layout, piping and instrumentation diagrams, pipe-rack and service-fluid load calculations, and bills of materials for a printing-ink manufacturing facility.

Junior Mechanical Engineer, **ME SOLshare Ltd.**

Mar 2019 – Sep 2019

Dhaka, Bangladesh

- Designed 3D-printed pogo-pin PCB calibration fixtures that reduced QA time by about 40 percent per board, iterating the design to cut material use for mass production.

- Engineered a modular tower for rural WiFi hardware that eliminated installation safety incidents, with zero reported in its first year, and developed robust enclosures for bi-directional solar meters.
- Authored field-deployment standard operating procedures for new hardware and software.

EDUCATION

Ph.D., Mechanical Engineering, University of Texas at Arlington *CGPA 3.86*

Aug 2022 – Dec 2026

B.Sc., Naval Architecture & Marine Engineering, Bangladesh University of Engineering and Technology (BUET)

2014 – 2018

TECHNICAL SKILLS

Additive Manufacturing: FFF/FDM, SLA, SLS, LPBF; Simplify3D, Cura, Stratasys Insight; Design for Additive Manufacturing (DfAM)

Materials Characterization: SEM, TEM, AFM, micro-CT, DSC/STA, FTIR, Raman, EDS, FLIR thermography, DIC, acoustic emission

CAD & Simulation: SOLIDWORKS (CSWA), AutoCAD, Rhino; Ansys, Abaqus, Altair Inspire, HyperMesh

Computational & Machine Learning: Python, MATLAB, C++, SAS; XGBoost, CatBoost, ANN, PCA/SVM; DoE, ANOVA, nested cross-validation

SELECTED PUBLICATIONS

Ahmed, R., Dola, I.S., Ahmed, S., et al. A review of design and fabrication strategies for thermoplastic lightweight thin-walled stiffened shells and lattice structures utilizing FFF. *Int. J. Advanced Manufacturing Technology*, 144:6111–6158, 2026.

Ahmed, R., Shanto, T.A., Rahman, M.M., Taylor, R.M., Jain, A. Machine learning based prediction of toughness in FFF: leveraging in-process annealing with enhanced printheads. *Solid Freeform Fabrication Symposium (SFF 2025)*, Austin, TX.

Patel, P., **Ahmed, R.**, Shanto, T.A., Jain, A., Taylor, R.M. Experimental characterization of enhanced FFF of tall thin-walled structures using PLA. *Int. J. Advanced Manufacturing Technology*, 139(11):5663–5675, 2025.

Patel, P., Rahman, M.M., Shanto, T.A., **Ahmed, R.**, et al. Evaluation of in process heating for extrusion printing of tall thin-walled structures. *Progress in Additive Manufacturing*, 2026.

Dola, I.S., **Ahmed, R.**, Zulqernine, M.J., Shanto, T.A., Taylor, R.M. Evaluation of deep learning architectures for defect detection in FFF. *SAMPE 2026*, Seattle, WA.

AWARDS & CERTIFICATIONS

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| • Dean's Doctoral Assistantship Award, UT Arlington | • SOLIDWORKS CAD Design Associate (CSWA); Member, ASME |
| • Dean's Fellowship Award, UT Arlington (Summer 2024 and 2026) | • Undergraduate Student Mentorship Award, UTA I-ENGAGE (2024) |
| • Lawrence Family Endowed Scholarship (2026) | • Certifications: GD&T, Programming for Everybody (Python) |
| • NSF Student Award, Solid Freeform Fabrication Symposium (2025) | • Developing Welding Procedure and Qualification |